



Machine Learning

AIML CZG565

M4 : Linear Models for Classification

Course Faculty of MTech Cluster

BITS – CSIS - WILP

BITS Pilani

Pilani Campus

Disclaimer and Acknowledgement



- These content of modules & context under topics are planned by the course owner Dr. Sugata, with grateful acknowledgement to many others who made their course materials freely available online
- We here by acknowledge all the contributors for their material and inputs.
- We have provided source information wherever necessary
- **Students are requested to refer to the textbook w.r.t detailed content of the presentation deck shared over canvas**
- **We have reduced the slides from canvas and modified the content flow to suit the requirements of the course and for ease of class presentation**

Slide Source / Preparation / Review:

From BITS Pilani WILP: Prof.Sugata, Prof.Chetana, Prof.Rajavadhana, Prof.Monali, Prof.Sangeetha, Prof.Swarna, Prof.Pankaj

External: CS109 and CS229 Stanford lecture notes, Dr.Andrew NG and many others who made their course materials freely available online

Agenda

- Discriminant Functions ✓
- Probabilistic Generative Classifiers ✓
- Probabilistic Discriminative Classifiers ✓
- Logistic Regression →
- Applications : Text classification model ✓

Decision Theory & Objective of Classification Models

Inductive Learning Hypothesis : Interpretation

- Target Concept : **t**
- Discrete : $f(x) \in \{\text{Yes, No, Maybe}\}$ Classification
- Continuous : $f(x) \in [20-100]$ Regression
- Probability Estimation : $f(x) \in [0-1]$



Sky	AirTemp	Humidity	Wind	Water	Forecast	<i>EnjoySport?</i>
Sunny	Warm	Normal	Strong	Warm	Same	Yes
Sunny	Warm	High	Strong	Warm	Same	Yes
Rainy	Cold	High	Strong	Warm	Change	No
Sunny	Warm	High	Strong	Cool	Change	Yes

Decision Theory

- Target Concept : t
- Discrete : $f(x) \in \{\text{Yes, No}\}$ ie., $t \in \{0, 1\}$ Binary Classification
- Continuous : $f(x) \in [20-100]$
- Probability Estimation : $f(x) \in [0-1]$

ML Task : Predict the Employability of interview candidates based on CGPA & IQ

Preprocess Implemented:

Min-Max Normalization on IQ

CGPA	IQ	IQ	Job Offered
5.5	6.7	100	1
5	7	105	0
8	6	90	1
9	7	105	1
6	8	120	0
7.5	7.3	110	0

Decision Theory :

The decision problem: given x , predict t according to a probabilistic model $p(x, t)$

- Target Concept : t
- Discrete : $f(x) \in \{\text{Yes, No}\}$ ie., $t \in \{0, 1\}$ ✓✓
- Continuous : $f(x) \in [20-100]$
- Probability Estimation : $f(x) \in [0-1]$

$$\begin{aligned}
 p \geq 0.5 &\rightarrow 1 \\
 p < 0.5 &\rightarrow 0 \\
 \hline
 0/1 \\
 0/1/2\dots
 \end{aligned}$$

$p(x, C_k)$ is the (central!) inference problem

↓ +ve - 1
-ve - 0

CGPA	IQ	IQ	Job Offered	P(Job = 1)
5.5	6.7	100	1	0.8
$X \leq 5$,	7 , >	105	0	<u>0.4</u> ?
8	6	90	1	0.75
9	7	105	1	0.95
6	8	120	0	0.35
7.5	7.3	110	0	0.4

$$P(C_k | x)$$

$$= P(C_k | x)$$

$$P(C_1 | x) = \underline{0.4}$$

$$\begin{aligned}
 &\downarrow \\
 &0.6 \\
 &7
 \end{aligned}$$

Classification Problem: Stages

$$p(C_k|x) = \frac{p(x|C_k)p(C_k)}{p(x)}$$

$$= \frac{p(x, C_k)}{p(x)} = \frac{p(x, C_k)}{\sum_{k=1}^2 p(x, C_k)}$$

C_k → discrimin.

CPGA	IQ	Job - Offered
5.5	6.7	1
5	7	0
8	6	1
9	7	1
6	8	0
7.5	7.3	0

Training Set

Induction/
Inference
step

Learning
algorithm

Learn
Model for
 $p(x, C_k)$

$p(x, C_{\text{job}=1})$ & $p(x, C_{\text{job}=0})$

Model

Apply Model
to find
optimal t

CPGA	IQ	Job - Offered
3	4	?
7	6	?
5.5	8	?

Test Set

Deduction/
Decision Step

$C_k \leftarrow$ discriminant
 $p(k) \leftarrow$ generative
classify

Decision Region

Sample Rule / Hypothesis:
 IF CGPA < 7 Job = 1
 Else Job = 0



Training Set

CPGA	IQ	Job - Offered
5.5	6.7	1
5	7	0
8	6	1
9	7	1
6	8	0
7.5	7.3	0

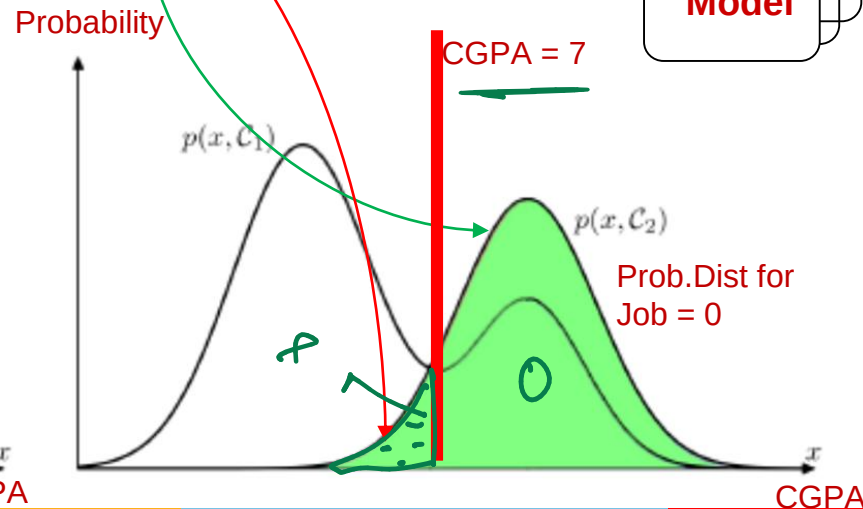
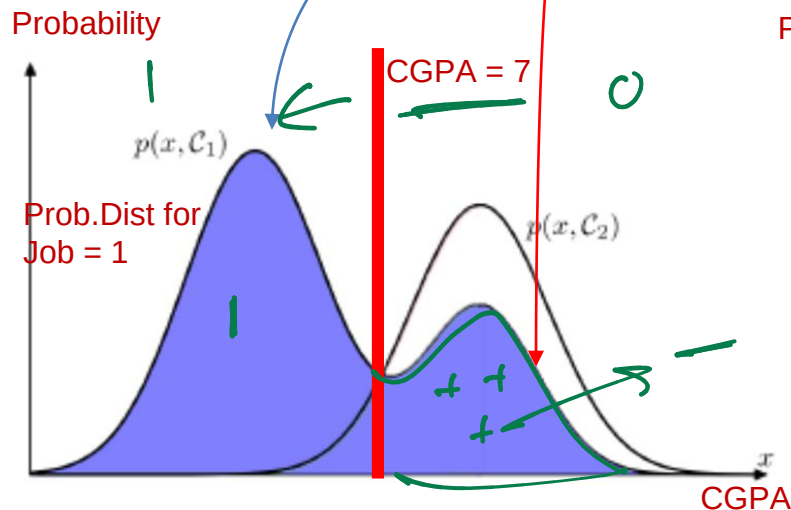
Learning algorithm

Learn Model for $p(x, C_k)$

Model divides the input space into regions R_k called **decision regions**, one for each class, such that all points in R_k are assigned to class C_k . A mistake occurs when an input vector belonging to class C_1 is assigned to class

Induction/
Inference
step

Model



Misclassification Rate

$$p(C_k|x) = \frac{p(x, C_k)}{p(x)}$$



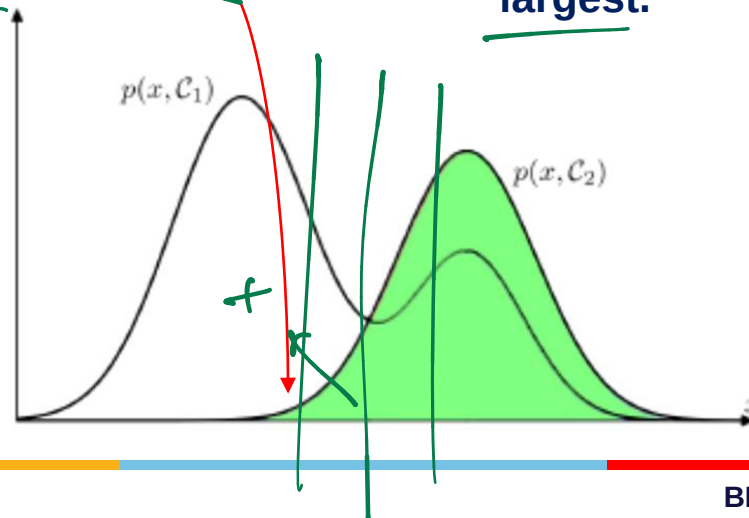
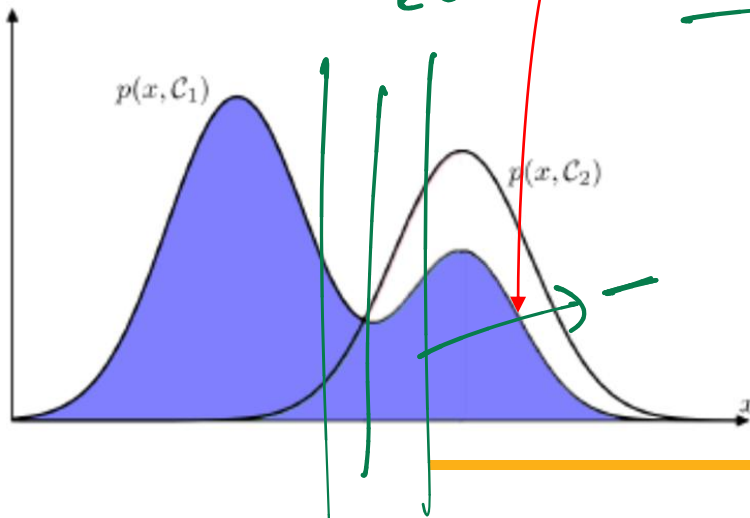
$$\begin{aligned} p(\text{mistake}) &= p(x \in \mathcal{R}_1, C_2) + p(x \in \mathcal{R}_2, C_1) \\ &= \int_{\mathcal{R}_1} p(x, C_2) dx + \int_{\mathcal{R}_2} p(x, C_1) dx. \end{aligned}$$

To minimize $p(\text{mistake})$, each x is assigned to whichever class has the smaller value of the integrand

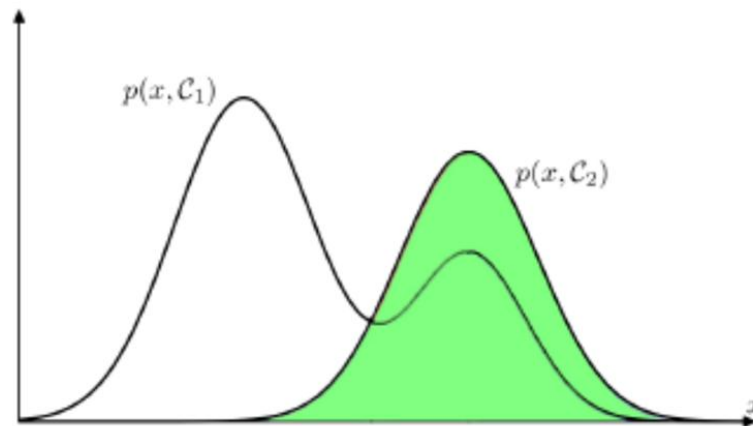
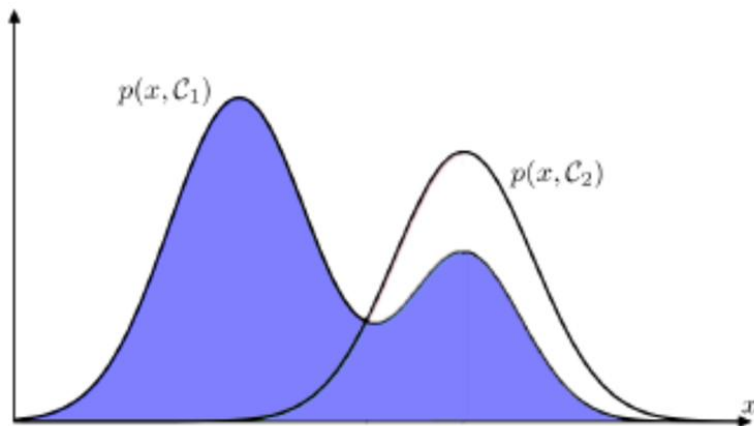
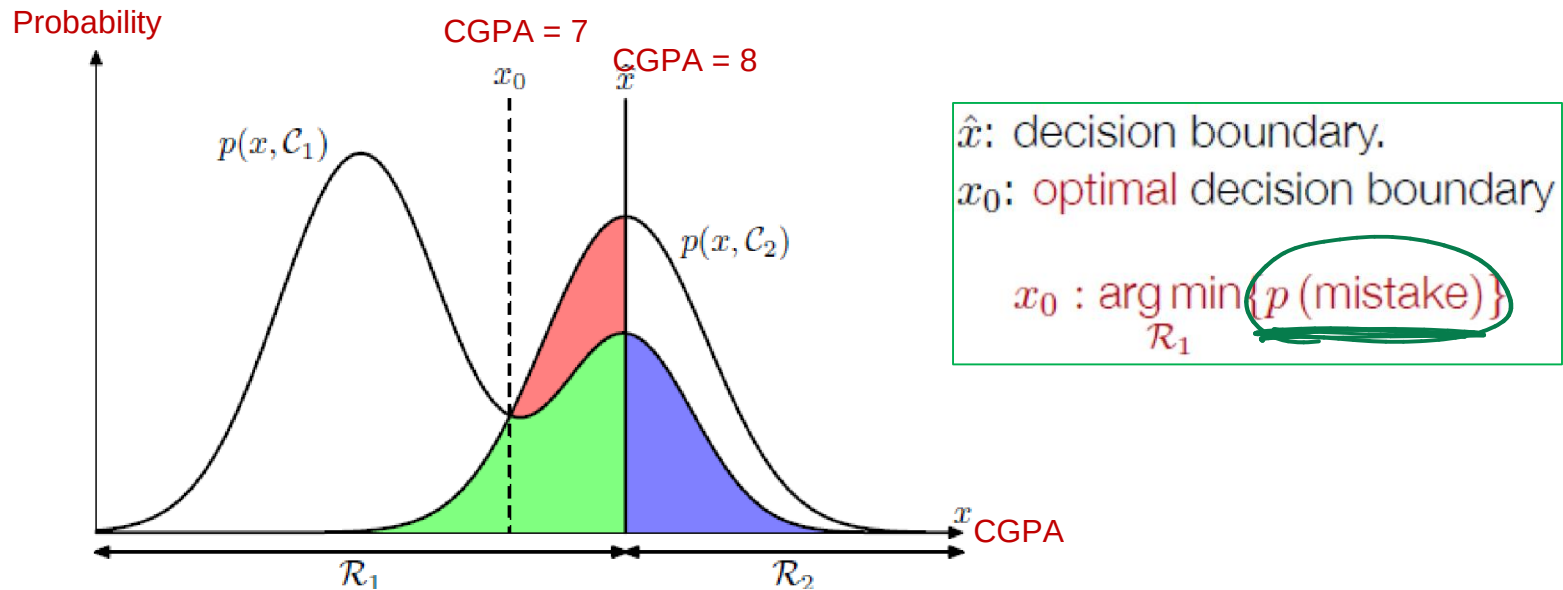
$$\begin{aligned} p(x, C_1) &> p(x, C_2) \\ \Leftrightarrow p(C_1|x)p(x) &> p(C_2|x)p(x) \\ \Leftrightarrow p(C_1|x) &> p(C_2|x) \end{aligned}$$

The minimum probability of making a mistake is obtained if each value of x is assigned to the class for which the **posterior probability** $p(C_k|x)$ is **largest**.

$$\min_{\text{error}} [p(C_1|x) \quad p(C_2|x)]$$



Decision Theory - Summary



Linear Models for Classification

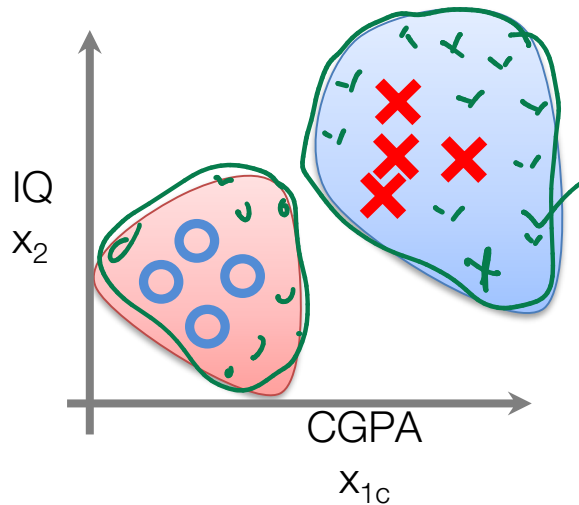
Types of Classification

Decision Theory: Interpretation



Building

Generative



$$P(Y | X_1 X_2 \dots X_n) = \frac{P(X_1 X_2 \dots X_d | Y) P(Y)}{P(X_1 X_2 \dots X_d)}$$

Known as generative models, because by sampling from them it is possible to generate synthetic data points in the input space.

Eg., Classification: **Naïve Bayes**, ✓
Clustering : **Mixtures of Gaussians**

$$P(c | x) = \frac{P(x | c) P(c)}{P(x)}$$

Likelihood: $P(x | c)$
Class Prior Probability: $P(c)$
Posterior Probability: $P(c | x)$
Predictor Prior Probability: $P(x)$

$$P(c | X) = P(x_1 | c) \times P(x_2 | c) \times \dots \times P(x_n | c) \times P(c)$$

CPGA	IQ	Job - Offered
5.5	6.7	1 ✗
5	7	0
8	6	1
9	7	1
6	8	0 ○
7.5	7.3	0
...		

Types of Classification

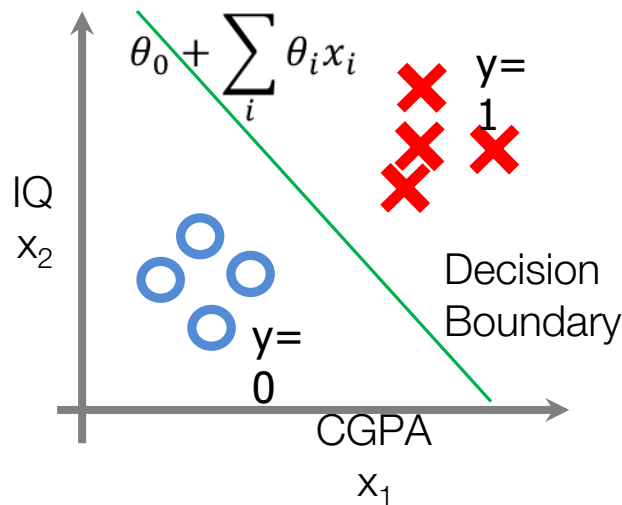
Decision Theory: Interpretation



Model

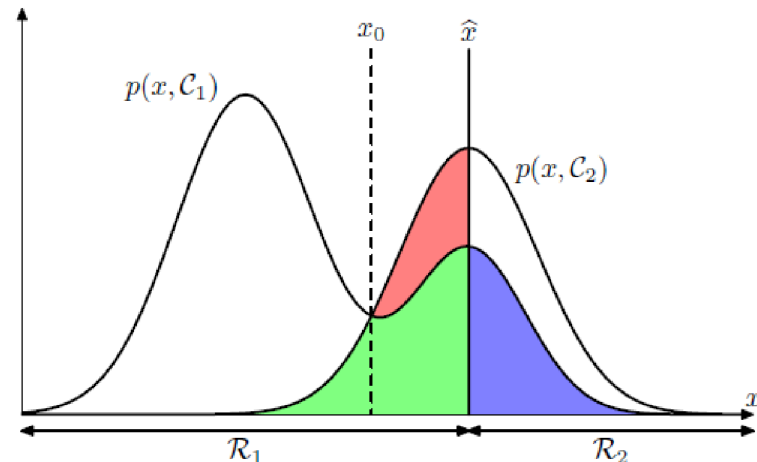
Building

Discriminative



$$\theta_0 + \sum_i \theta_i x_i \geq 0$$
$$\theta_0 + \sum_i \theta_i x_i < 0$$

✓ Logistic regression, SVMs, tree based classifiers (e.g. decision tree) Traditional neural networks, Nearest neighbor ✓



$$P(c | x) :$$

Posterior Probability

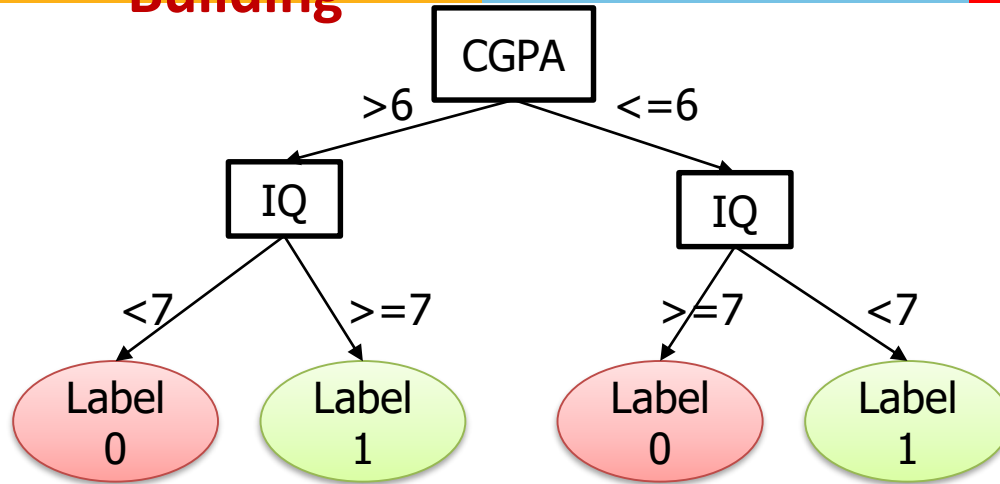
CPGA	IQ	Job - Offered
5.5	6.7	1
5	7	0
8	6	1
9	7	1
6	8	0
7.5	7.3	0
...		

Types of Classification

Decision Theory: Interpretation



Building



IF CGPA > 6 and IQ $\geq$ 7

Job offered = 1

Else If CGPA $\leq$ 6 and IQ < 7

Job offered = 1

Else if CGPA $\leq$ 6 and IQ $\geq$ 7

Job offered = 0

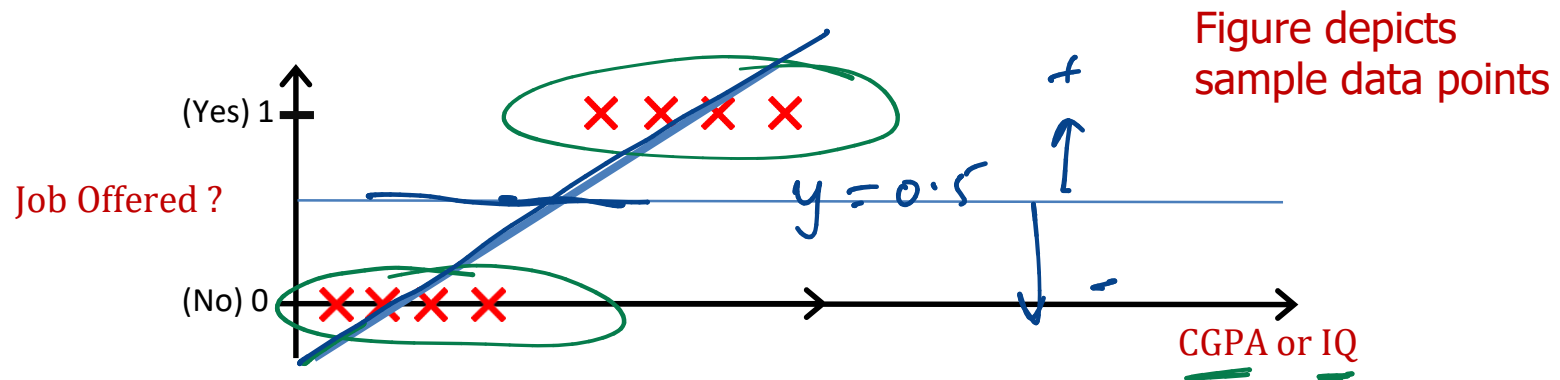
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Logistic regression, SVMs , tree based classifiers (e.g. decision tree) Traditional neural networks, **Nearest neighbor**

CPGA	IQ	Job - Offered
5.5	6.7	1
5	7	0
8	6	1
9	7	1
6	8	0
7.5	7.3	0
...		

Logistic Regression

Logistic Regression vs Least Squares Regression



- Independent Attribute : CGPA or IQ
- Can we solve the problem using linear regression? E.g., fit a straight line and define a threshold at 0.5
- Threshold classifier output $h_{\theta}(x)$ at 0.5:

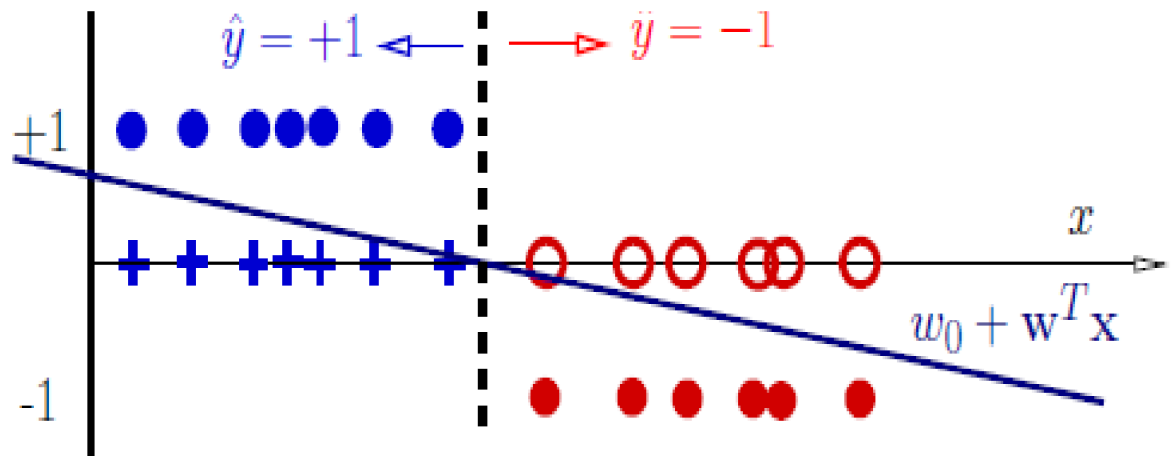
A Discriminant function $f(x)$ directly map input to class labels
In two-class problem, $f(.)$ is binary valued

If $h_{\theta}(x) \geq 0.5$, predict “ $y = 1$ ”

If $h_{\theta}(x) < 0.5$, predict “ $y = 0$ ”

In this use case : $h_{\theta}(x) = 0.7$, implies that there is 70% of chance of the candidate being selected in the interview

Decision Rules



- Classifier:

$$f(\mathbf{x}, \mathbf{w}) = w_0 + \mathbf{w}^T \mathbf{x} \quad (\text{linear discriminant function})$$

LDF

- Decision rule is

$$y = \begin{cases} 1 & \text{if } f(\mathbf{x}, \mathbf{w}) \geq 0 \\ -1 & \text{otherwise} \end{cases}$$

- Mathematically

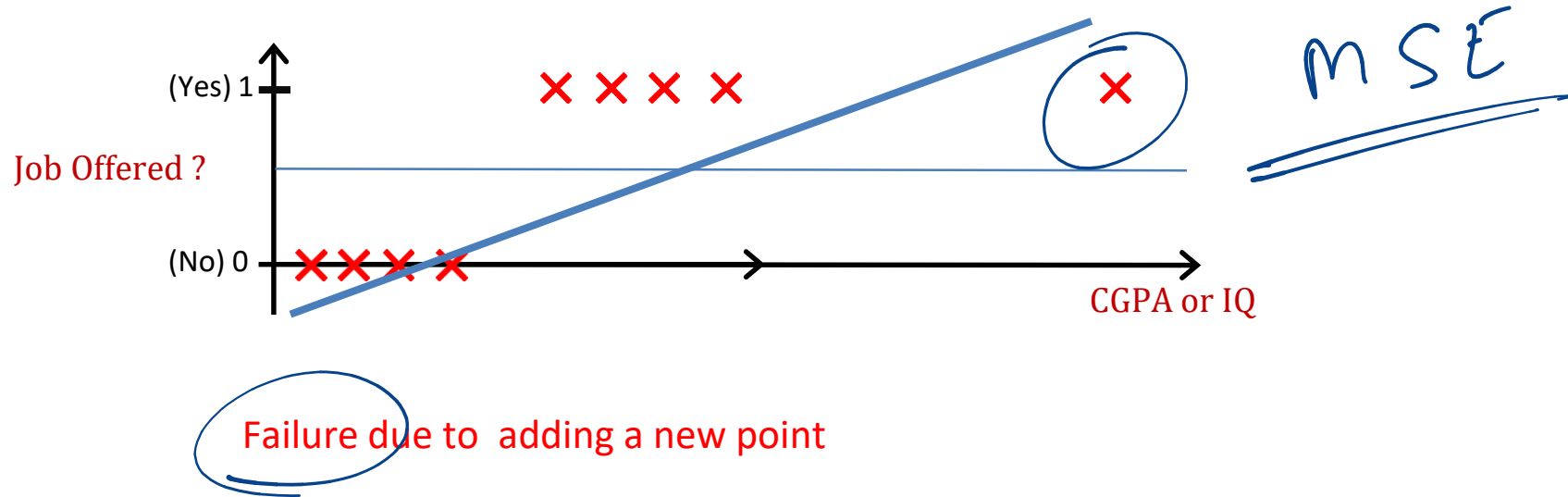
$$y = \text{sign}(w_0 + \mathbf{w}^T \mathbf{x})$$

- This specifies a **linear classifier**: it has a **linear boundary (hyperplane)**

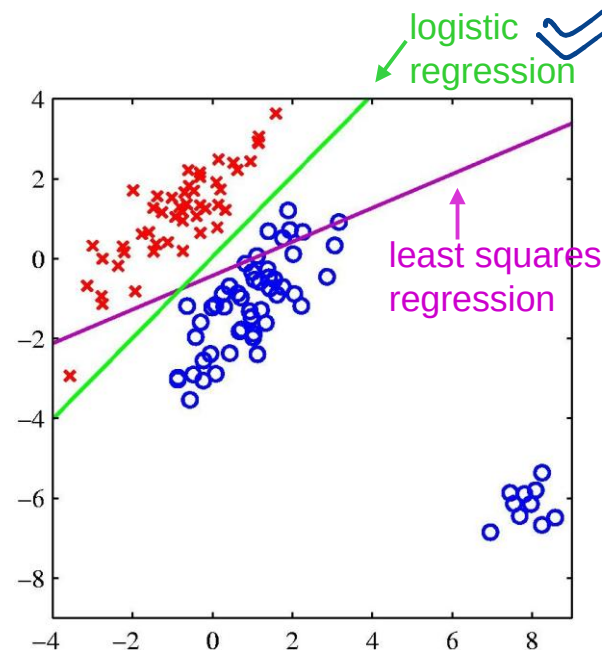
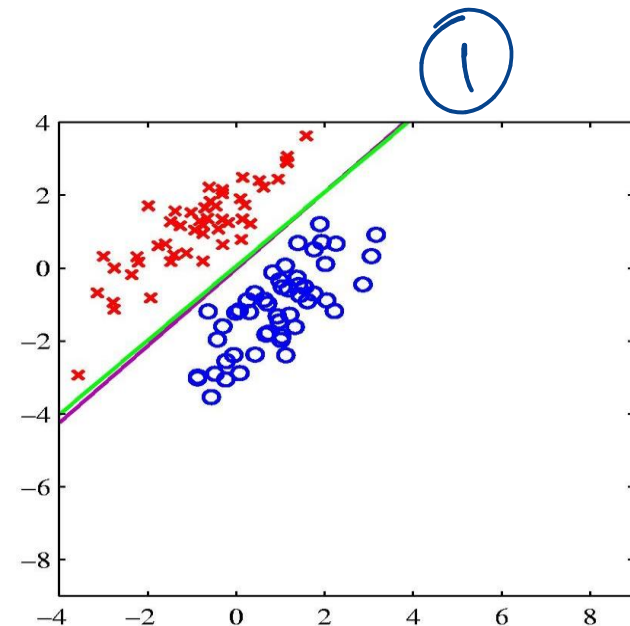
$$w_0 + \mathbf{w}^T \mathbf{x} = 0$$

A discriminant is a function that takes an input vector $\mathbf{x}$ and assigns it to one of K classes, denoted C_k .

Logistic Regression vs Least Squares Regression



Logistic Regression vs Least Squares Regression



The right-hand plot shows the corresponding results obtained when extra data points are added at the bottom left of the diagram, showing that least squares is highly sensitive to outliers, unlike logistic regression.

Logistic Sigmoid Function

→ Reg?

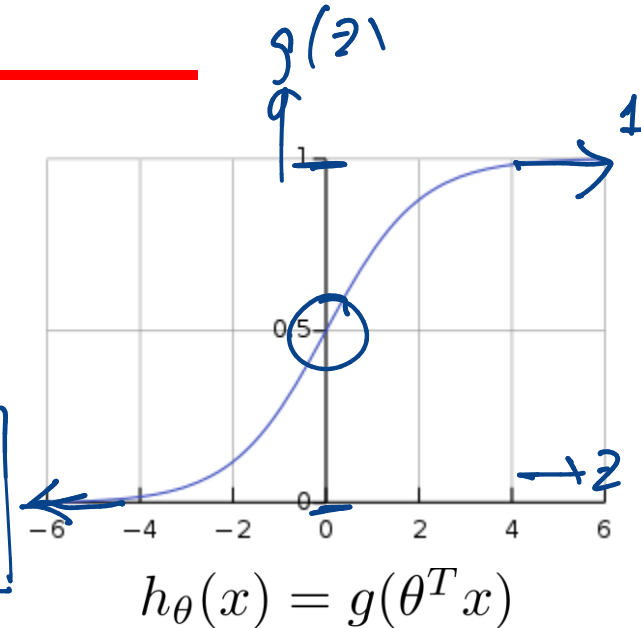


- Sigmoid/logistic function takes a real value as input and outputs another value between 0 and 1
- That framework is called logistic regression
 - Logistic: A special mathematical sigmoid function it uses
 - Regression: Combines a weight vector with observations to create an answer

• Want $0 \leq h_{\theta}(x) \leq 1$

• $h_{\theta}(x) = g(\theta^T x)$,

where $g(z) = \frac{1}{1+e^{-z}}$



• Sigmoid function

• Logistic function

$$h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}}$$

$$z \rightarrow (-\infty) - (+\infty)$$

$$\downarrow$$

$$g(z) \rightarrow 0 - 1$$

Lin Reg $h_0(x) = \theta^T x$ $W^T x$ What is θ

$-\infty$ $+\infty$

$\frac{\theta^T x}{e} \rightarrow \text{line}$

Log. Reg = $h_\theta(x)$ (0-1) + non linear

$$= \sigma(\theta^T x) = \frac{1}{1 + e^{-\theta^T x}} = p$$

D Boundary

linear $\leftarrow \theta^T x$

$\frac{0-1}{p}$ $\ln \frac{p}{1-p}$ log odds ratio

$- \infty$ $-\infty$ $+\infty$



Logistic Regression vs Least Squares Regression

Classification: $y = 0$ or 1



$h_{\theta}(x)$ can be > 1 or < 0

Another drawback
of using linear
regression for this
problem

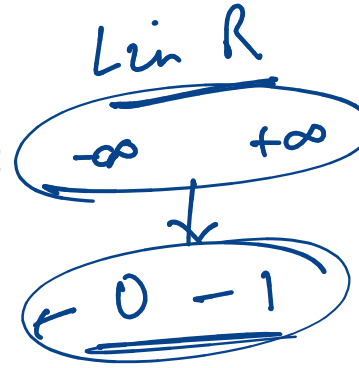
What we need: $0 \leq h_{\theta}(x) \leq 1$

Logistic Regression:



$P(1) = 0 \rightarrow \text{cl } 0$
 $P(1) = 1 \rightarrow \text{cl } 1$

$$Y = \theta_0 + \sum_i \theta_i x_i$$



$$p = \frac{1}{1 + e^{-y}}$$

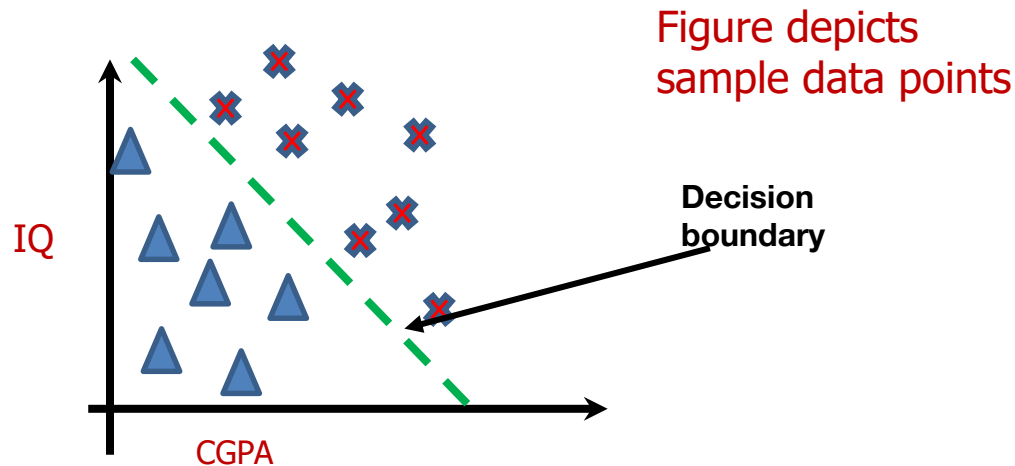
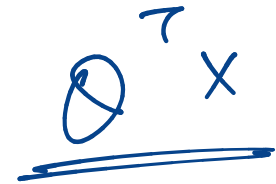
$$h_{\theta}(x) = g(\theta^T x)$$

$$\ln \frac{p}{1-p} = \theta_0 + \sum_i \theta_i x_i$$

Logistic Regression – Sample Linear Boundary



- At decision boundary output of logistic regression is 0.5
- $h_{\theta}(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$
 - e.g., $\theta_0 = -3, \theta_1 = 1, \theta_2 = 1$



- Predict “ $y = 1$ ” if $-3 + x_1 + x_2 \geq 0$

Logistic Regression – Sample Non-Linear Boundary

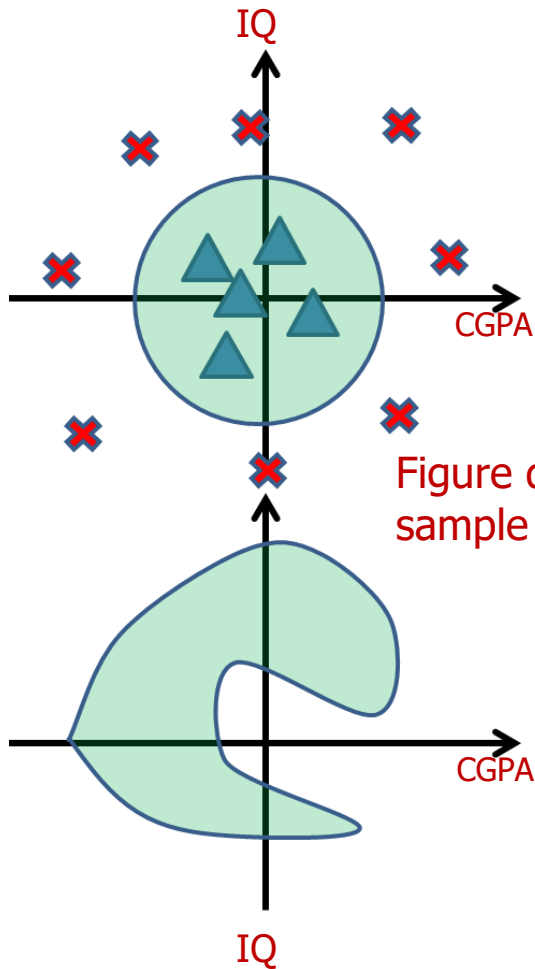


Figure depicts
sample data points

- $h_{\theta}(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2 + \theta_4 x_2^2)$
E.g., $\theta_0 = -1, \theta_1 = 0, \theta_2 = 0, \theta_3 = 1, \theta_4 = 1$
- Predict “ $y = 1$ ” if $-1 + x_1^2 + x_2^2 \geq 0$

- $h_{\theta}(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2 + \theta_4 x_1^2 x_2 + \theta_5 x_1^2 x_2^2 + \theta_6 x_1^3 x_2 + \dots)$

SVM
ANN

Learning model parameters

- Training set:

$$\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \dots, (x^{(m)}, y^{(m)})\} \quad \checkmark$$

- m examples

$$x \in \begin{bmatrix} x_0 \\ x_1 \\ \dots \\ x_n \end{bmatrix} \quad x_0 = 1, y \in \{0, 1\}$$

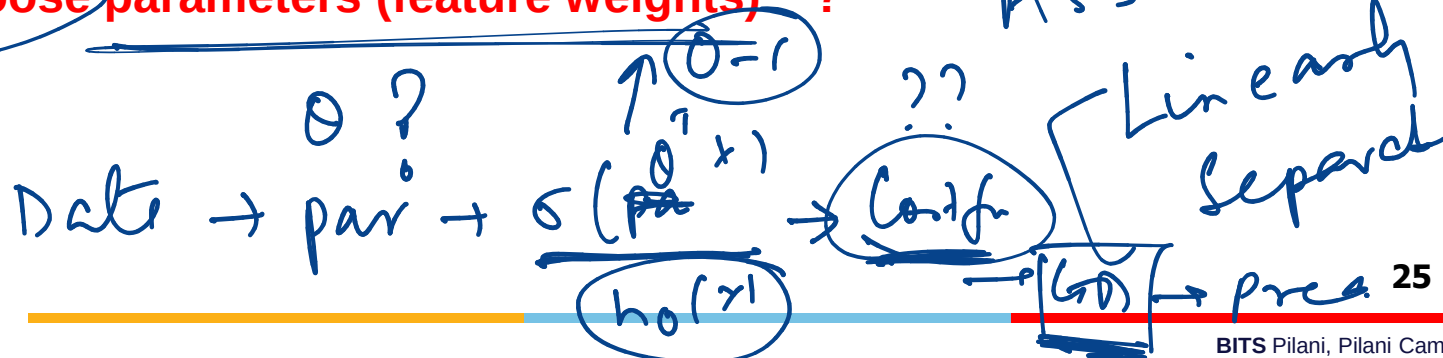
23rd May
Holidays
ML

→ class for x

$$\sigma(\theta^T x) \quad \boxed{h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}}}$$

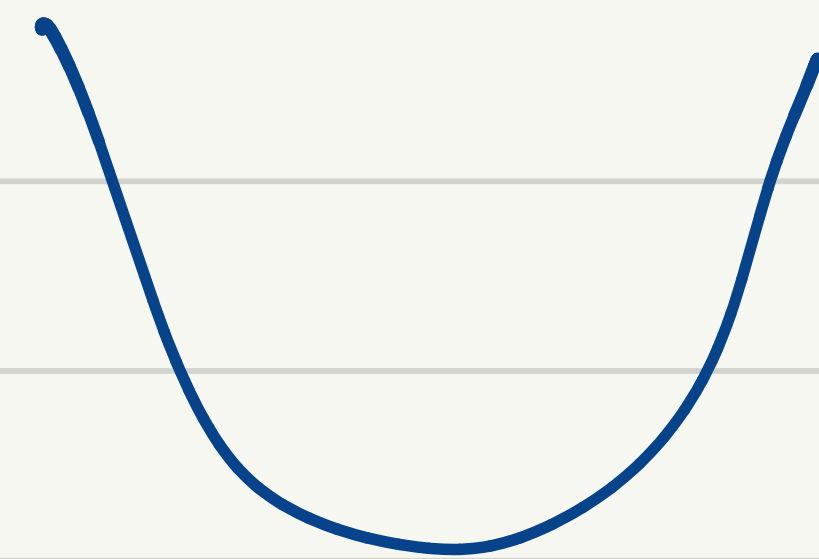
- How to choose parameters (feature weights) ?

Assume:
Linearly Separable



$$J(\theta) = (\quad)^2 + (\quad)^2$$

Linear Rep.



~~h₀(x)~~ $\checkmark$

$$h_0(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 \dots + \theta_d x_d$$

$$\theta_0 + \sum_{j=1}^d \theta_j x_j$$

bias

$$\left[\theta_0 \quad \theta_1 \quad \theta_2 \quad \dots \quad \theta_d \right] \cdot \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ \dots \\ x_d \end{bmatrix}$$

$$\left[x_0 = 1 \right]$$

$$\theta^T X$$

(

$$\begin{bmatrix} \theta_0 \\ \vdots \\ \theta_d \end{bmatrix}$$

2

+

sign

$$h_0(x) = \sum_{j=0}^d \theta_j x_j$$

d+1

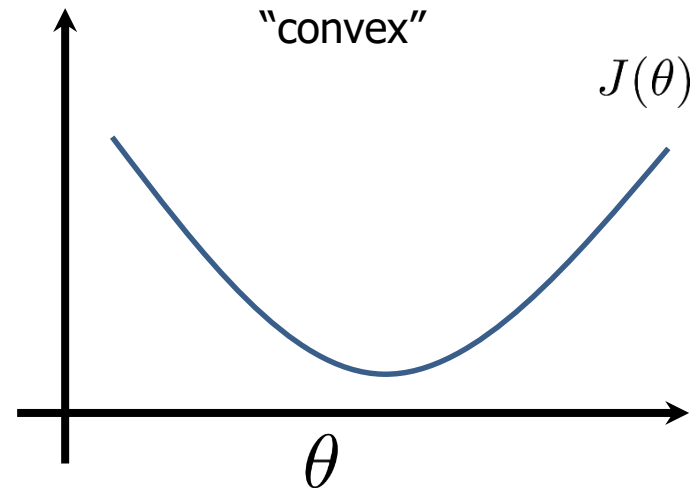
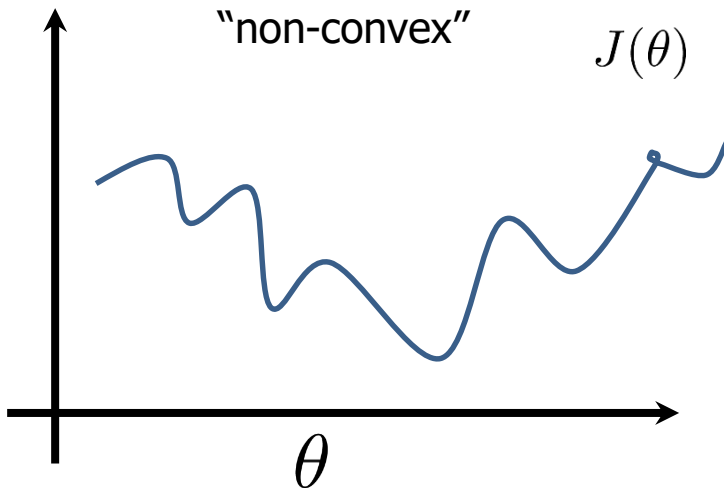
$$\begin{bmatrix} 1 \\ x_1 \\ x_2 \\ \dots \\ x_d \end{bmatrix}$$

augmented
feature vector

Notion of Cost Function in Classification

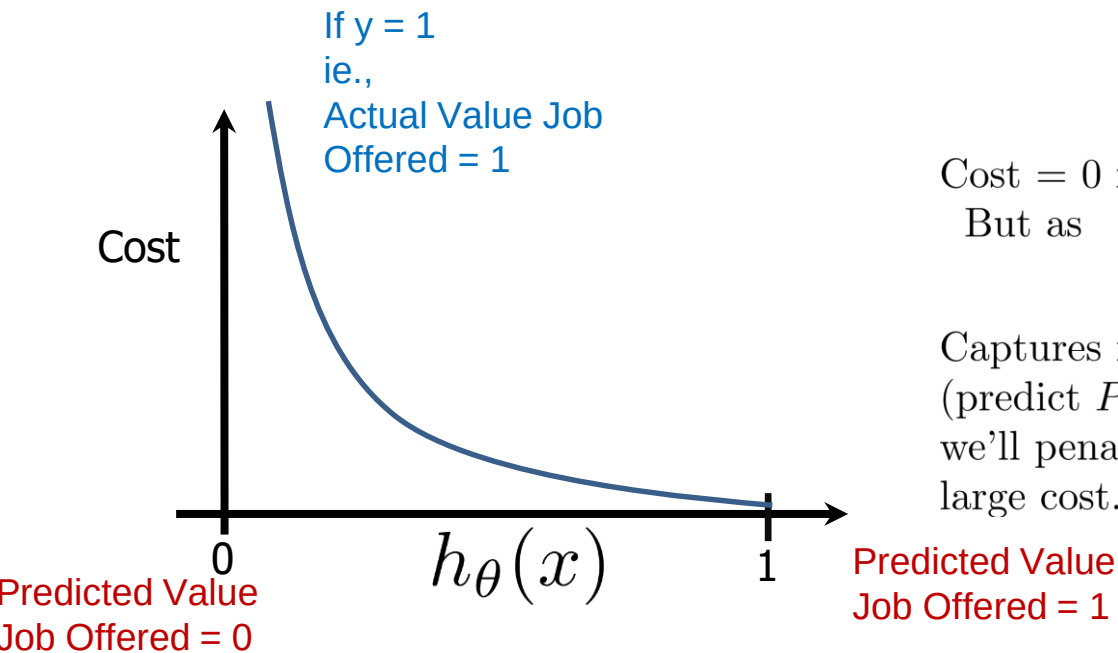
Logistic Regression

- Training set: $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \dots, (x^{(m)}, y^{(m)})\}$
- How to choose parameters (feature weights)? θ



Logistic regression cost function (cross entropy)

$$\text{Cost}(h_{\theta}(x), y) = \begin{cases} -\log(h_{\theta}(x)) & \text{if } y = 1 \\ -\log(1 - h_{\theta}(x)) & \text{if } y = 0 \end{cases}$$



Cost = 0 if $y = 1, h_{\theta}(x) = 1$

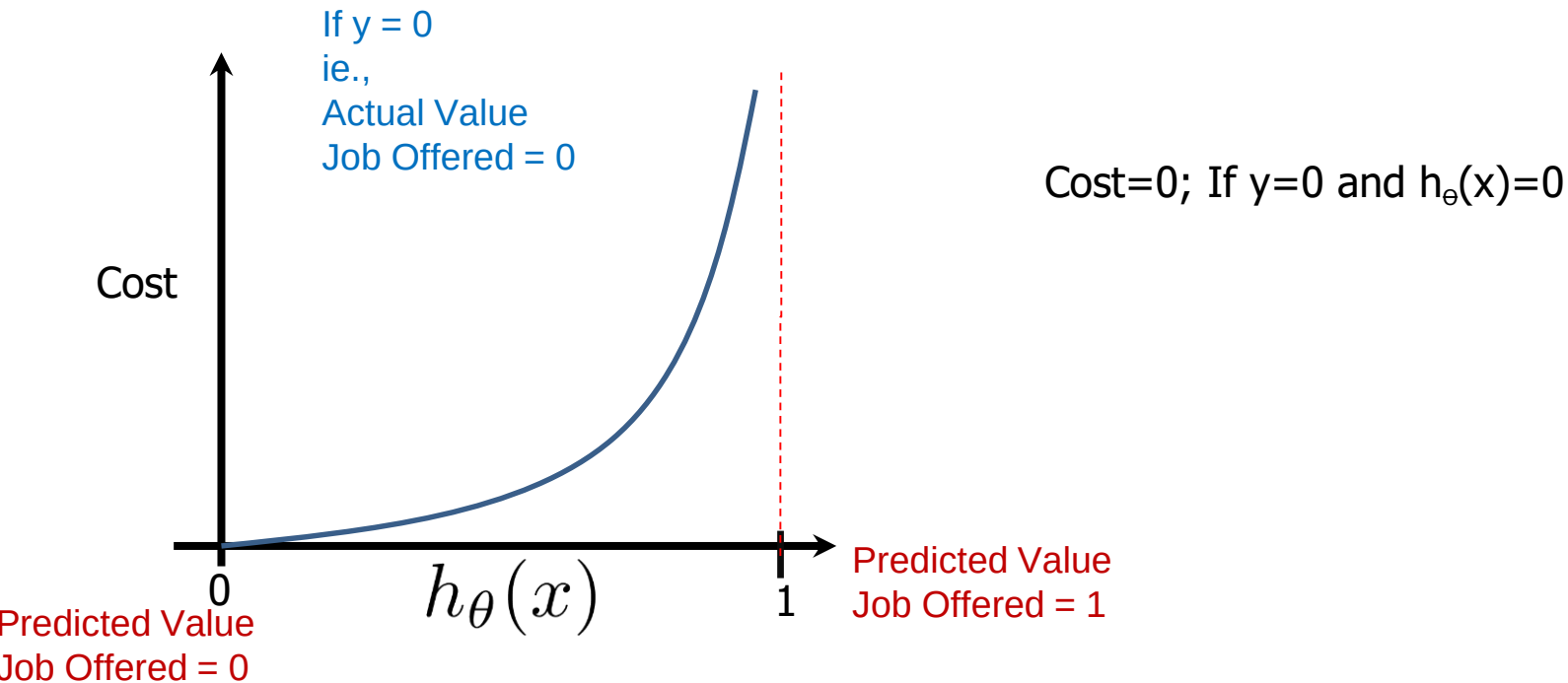
But as $h_{\theta}(x) \rightarrow 0$

$Cost \rightarrow \infty$

Captures intuition that if $h_{\theta}(x) = 0$,
(predict $P(y = 1|x; \theta) = 0$), but $y = 1$,
we'll penalize learning algorithm by a very
large cost.

Logistic regression cost function

$$\text{Cost}(h_{\theta}(x), y) = \begin{cases} -\log(h_{\theta}(x)) & \text{if } y = 1 \\ -\log(1 - h_{\theta}(x)) & \text{if } y = 0 \end{cases}$$



Cost function

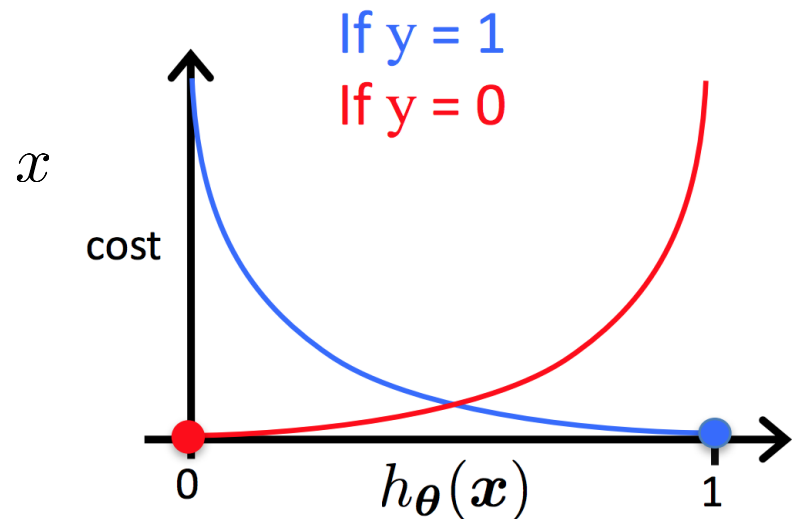
$$J(\theta) = \frac{1}{m} \sum_{i=1}^m \text{Cost}(h_{\theta}(x^{(i)}), y^{(i)})$$

$$= -\frac{1}{m} \left[\sum_{i=1}^m y^{(i)} \log h_{\theta}(x^{(i)}) + (1 - y^{(i)}) \log (1 - h_{\theta}(x^{(i)})) \right]$$

To fit parameters θ : Apply Gradient Descent Algorithm $\min_{\theta} J(\theta)$

To make a prediction given new :

Output :
$$h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}}$$



Gradient Descent Algorithm

$$J(\theta) = -\frac{1}{m} \left[\sum_{i=1}^m y^{(i)} \log(h_{\theta}(x^{(i)})) + (1 - y^{(i)}) \log(1 - h_{\theta}(x^{(i)})) \right]$$

Goal: $\min_{\theta} J(\theta)$

Repeat

{

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta)$$

}

$$\frac{\partial}{\partial \theta_j} J(\theta) = \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) x_j^{(i)}$$

Gradient Descent Algorithm

Linear Regression

• Repeat {

$$\theta_j := \theta_j - \alpha \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) x_j^{(i)}$$

 }

$$h_{\theta}(x) = \theta^{\top} x$$

Logistic Regression

Repeat {

$$\theta_j := \theta_j - \alpha \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) x_j^{(i)}$$

 }

$$h_{\theta}(x) = \frac{1}{1 + e^{-\theta^{\top} x}}$$

$$h_{\theta}(x) = \frac{1}{1 + e^{-(\theta_0 + \theta_1 CGPA + \theta_2 IQ)}}$$

Slide credit: Andrew
Ng

Thank you !

Required Reading for completed session :

T1 - Chapter # 6 (Tom M. Mitchell, Machine Learning)

R1 – Chapter # 3,#4 (Christopher M. Bhisop, Pattern Recognition & Machine Learning) & Refresh your MFDS course basics